
Bounds of Cycle Integrals of Modular Forms

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1. Introduction and Background

Motivated by the classical theory of complex multiplication, Kaneko introduced the concept of evaluating the modular j -function at real quadratic irrationalities. This evaluation is defined as the integral of the function over the corresponding closed geodesics. Kaneko hypothesized that these specific cycle integrals yield explicitly bounded values. Recently, significant progress has been made on this front; it was proven that for the j -function, the real part of the value for any quadratic irrational w is bounded below by the value at the golden ratio, denoted as ϕ . Specifically, $\operatorname{Re}(val_j(w)) \geq \operatorname{Re}(val_j(\phi))$. This project aims to generalize Kaneko's Conjecture and explore its extensions to modular forms beyond the j -function. By leveraging experimental mathematics—specifically using the computational software SAGE—we tabulate cycle integrals of various modular forms, formulate analogous conjectures, and attempt to prove them using strategies similar to those employed for the j -function.

1.1 The Modular Framework

To formalize this, we consider the group $SL_2(\mathbb{Z})$, which consists of 2×2 matrices with integer entries and a determinant of 1. This group acts on the upper-half plane of complex numbers via fractional linear transformations:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} z = \frac{az + b}{cz + d}.$$

A modular form of weight k is a holomorphic function on the upper-half plane that satisfies specific transformation properties under the action of $SL_2(\mathbb{Z})$. Specifically, it satisfies the relation:

$$f\left(\frac{az + b}{cz + d}\right) = (cz + d)^k f(z),$$

along with additional boundedness conditions. Klein's modular invariant, $j(\tau)$, is a prominent example of a weakly holomorphic modular form of weight 0.

2. Values of Real Quadratic Irrationals

While modular forms are strictly defined on the upper-half complex plane, it is mathematically rich to evaluate their “values” at real quadratic irrationals. Let w be a real quadratic irrational. It acts as a fixed point for a specific matrix transformation $A \in SL_2(\mathbb{Z})$. From this matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, we can extract a quadratic form $Q(x, y) = cx^2 + (d - a)xy - by^2$ (up to a sign) characterized by a discriminant D . For the classical j -function, the value at w is defined via the cycle integral:

$$val_j(w) = \frac{1}{2 \ln(\epsilon_w)} \int_{\tau}^{A\tau} \frac{j(z)\sqrt{D}}{Q(z, 1)} dz,$$

where τ is an arbitrary point in the upper half plane, and ϵ_w represents the largest eigenvalue of the matrix A .

2.1 Extension to Arbitrary Weights

To extend this framework, we define the value of a quadratic irrational w for an arbitrary modular form f of weight $-2n$ as:

$$val_f(w) = \frac{1}{2 \ln(\epsilon_w)} \int_{\tau}^{A\tau} f(z) \left(\frac{\sqrt{D}}{Q(z, 1)} \right)^{n+1} dz.$$

Because f is constrained to have a weight of $-2n$, this integral is well-defined and remains independent of the choice of the base point τ .

3. Computational Methodology

To facilitate large-scale experimental testing, we required a computational approach to evaluate $val_f(w)$. Previous literature derived a formula for val_j making it conducive to computer calculation. We successfully extended this formulation to encompass general modular forms of weight $-2n$.

3.1 Algorithmic Implementation The computational strategy relies on the continued fraction expansions of real quadratic irrationals.

- Every real quadratic irrational possesses a purely periodic continued fraction in its orbit under the action of $SL_2(\mathbb{Z})$.
- Because the value of a quadratic irrational is invariant under $SL_2(\mathbb{Z})$, we can deliberately select this periodic continued fraction expression to simplify calculations.
- A purely periodic continued fraction $w = \overline{[a_1, \dots, a_{2k}]}$ is a fixed point of a matrix constructed from products of matrices T and V , where $T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $V = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$. Using this representation, we developed an algorithmic method to generate two sequences, $w_{k,m}$ and $\overline{w}_{k,m}$, which allows us to rewrite the integral over the unit circle parameterization $z = e^{it}$:

$$val_f(w) = \int_{\pi/3}^{2\pi/3} f(e^{it}) i e^{-nit} \sum_{k=1}^{2l} \sum_{m=1}^{a_k} \left(\frac{e^{it}}{e^{it} - w_{k,m}} - \frac{e^{it}}{e^{it} - \overline{w}_{k,m}} \right)^{n+1} dt.$$

4. Experimental Results

We tested our algorithmic approach using Eisenstein series, defined as $E_{2k}(\tau) = \frac{1}{2\zeta(2k)} \sum'_{m,n} \frac{1}{(m\tau+n)^{2k}}$ over all pairs of integers except $(0,0)$, and the modular discriminant $\Delta = \frac{(2\pi)^{12}}{1728} (E_4^3 - E_6^2)$.

4.1 Testing Weight -2: $f = \frac{E_4 E_6}{\Delta}$ Our first test case was the modular form $f = \frac{E_4 E_6}{\Delta}$, which has a weight of -2. Numerical calculations yielded $Re(val_f(\phi)) \approx -3.81 \cdot 10^{-14}$ and $Re(val_f(\overline{[2, 2]})) \approx -3.58 \cdot 10^{-14}$. It can be rigorously proven that the true theoretical values are $Re(val_f(\phi)) = 0$ and $Re(val_f(\overline{[2, 2]})) = 0$, indicating that our numerical algorithm produces outputs very close to the true values of these quadratic irrationals. Further calculations showed $Re(val_f(\overline{[1, 2]})) \approx -26.26$ and $Re(val_f(\overline{[4, 1]})) \approx 34.72$. Conclusion for Weight -2: These varying positive and negative results demonstrate that for $f = \frac{E_4 E_6}{\Delta}$, the value no longer exhibits a clear minimum or maximum.

4.2 Testing Weight 0: $f = \frac{E_4^2}{\Delta}$

We subsequently analyzed the modular form $f = \frac{E_4^2}{\Delta}$. We tabulated the following numerical results:

$$\begin{aligned} \phi &\mapsto 99.93 \\ \overline{[1, 2]} &\mapsto 62.47 \\ \overline{[4, 2]} &\mapsto 5.494 \\ \overline{[1, 2, 3, 4]} &\mapsto 14.99 \\ \overline{[5, 1, 3, 2]} &\mapsto 12.18 \\ \overline{[100, 1]} &\mapsto -19.94 \end{aligned}$$

Conclusion for $f = \frac{E_4^2}{\Delta}$: This pattern of values was consistent across all checked quadratic irrationals. The data strongly suggests that $Re(val_f(\phi)) \approx 99.93$ serves as an absolute upper bound for the values of quadratic irrationals evaluated under this specific modular form.

5. Analytical Framework and Proof Strategy

To rigorously prove the upper bound observed for $f = \frac{E_4^2}{\Delta}$, we translated the problem into the study of auxiliary integral functions.

5.1 A Useful Lemma

For a modular form f of weight $-2n$ (where n is even), the function $e^{-nit} f(e^{it})$ is real-valued on the interval $t \in [\frac{\pi}{3}, \frac{2\pi}{3}]$. This allows us to express the real part of the value elegantly as:

$$Re(val_f(w)) = \int_{\pi/3}^{2\pi/3} e^{-nit} f(e^{it}) S_w^n(t) dt$$

Here, $S_w^n(t)$ is an auxiliary function derived from the continued fraction sequences. For even n , $S_w^n(t)$ exhibits symmetry $S_w^n(\pi - t) = S_w^n(t)$. To compare the value at the golden ratio ϕ against an arbitrary irrational w , we define a difference function $D_{\phi,w}^n(t) = S_\phi^n(t) - S_w^n(t)$. We can then express the difference in their evaluations as:

$$Re(val_f(\phi)) - Re(val_f(w)) = \int_{\pi/3}^{2\pi/3} e^{-nit} f(e^{it}) D_{\phi,w}^n(t) dt.$$

If we can mathematically guarantee that this integral is strictly non-negative for any w , it rigorously establishes that ϕ provides a maximal value.

5.2 Integral Inequality Bounds To evaluate the positivity of the integral, we rely on a modified analytical lemma regarding continuous functions: Let $D(t)$ be a continuous function where $D(\pi - t) = D(t)$, D is decreasing on the interval, and $\int_{\pi/3}^{2\pi/3} D(t) dt = 0$. Let $g(t)$ be a continuous function satisfying $g(\pi - t) = g(t)$. If both g and D are non-positive and decreasing on the relevant interval, then the integral $\int_{\pi/3}^{2\pi/3} g(t) D(t) dt$ is greater than or equal to 0.

6. Extending the Lemma to $f = \frac{E_4^2}{\Delta}$ We attempted to apply this analytical lemma to our experimental success case by setting $D(t) = D_{\phi,w}^n(t)$ and $g(t) = e^{-nit} \frac{E_4^2(e^{it})}{\Delta(e^{it})}$. A significant challenge is that, except for $n = 0$, the condition $\int_{\pi/3}^{2\pi/3} D_{\phi,w}^n(t) dt = 0$ does not universally hold. However, numerical graphing strongly suggests that our target function $g(t)$ is strictly non-positive on $[\frac{\pi}{3}, \frac{2\pi}{3}]$. To definitively prove that $g(t)$ is decreasing (a core requirement of the lemma), we investigated its logarithmic derivative. Utilizing Ramanujan's formulas for the derivatives of Eisenstein series, we computed:

$$\frac{g'(t)}{g(t)} = -nit + \frac{2\pi}{3} e^{it} [E_2(e^{it}) + 2 \frac{E_6(e^{it})}{E_4(e^{it})}].$$

Consequently, $g(t)$ will satisfy all required conditions to prove the upper bound if we can demonstrate that:

$$Re \left(e^{it} \left[E_2(e^{it}) + 2 \frac{E_6(e^{it})}{E_4(e^{it})} \right] \right) \geq 0$$

for $t \in [\frac{\pi}{3}, \frac{2\pi}{3}]$.

7. Unfinished Work and Remaining Questions

While our experimental data strongly supports the existence of bounds for certain modular forms beyond the j -function, formal mathematical proof requires overcoming specific analytical hurdles. Moving forward, our team seeks to resolve the following questions.

- **Rigorous Proof of Monotonicity:** What specific analytical techniques are required to complete the rigorous proof that $g(t)$ is strictly decreasing on the required interval?
- **Generalization to Odd Weights:** The current symmetrical properties rely heavily on n being even. How can we refine this conjecture to accommodate modular forms where n is odd?
- **Broader Application:** Does this framework and associated lemma apply universally to other modular forms of weight $-2n$ (for even n)? More haphazard investigation of the modular form E_8 suggests that ϕ might return to being a lower bound in this case, and for $\frac{E_6}{\Delta}$ ϕ doesn't seem to be a minimum or maximum.